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KENBRIGHT ACTUARIAL AND FINANCIAL SERVICES LIMITED

CAPITAL ADEQUACY ANALYSIS

INTERNAL TECHNICAL PROCEDURES

VERSION 1.0

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KAFS ITP

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Kenbright Actuarial & Financial Services Limited
26th Floor- Prism Towers – 3rd Ngong Avenue
P. O. Box 28281 – 00200
Nairobi, Kenya
kafs@kenbright.co.ke

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Prepared for:

Kenbright Actuarial & Financial Services Limited
26th Floor, Prism Towers
3rd Ngong Avenue, Upperhill
P.O Box 28281 - 00200
Nairobi, Kenya

OUR REF

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DOCUMENT PURPOSE

This document has been produced as part of the internal technical procedures for KAFS and provides guidance on capital adequacy analysis for general and life insurance businesses.

DOCUMENT OWNERSHIP

The owner of this document is Kenbright Actuarial and Financial Services Limited

TABLE OF CONTENTS

1 INTRODUCTION 5
Purpose 5
Key Concepts 5
2 CAPITAL ADEQUACY CALCULATION 6
Available Capital 6
Tier 1 Capital 6
Tier 2 Capital 7
Deductions (Inadmissible Assets) 7
Required Capital (Risk Based Capital) 9
Insurance Risk 9
Catastrophe Risk 9
Market Risk 10
Credit Risk 10
Operational Risk 10
3 CAR TEMPLATE 12
4 SUPERVISORY ACTIONS 16
5 APPENDICES 17
Appendix 1 17
Appendix 2 18
Appendix 3 19
Appendix 4 19

ABBREVIATIONS

NAME	DEFINITION
BEL	Best Estimate Liability
CAR	Capital Adequacy Ratio
IRA	Insurance Regulatory Authority
KAFS	Kenbright Actuarial & Financial Services Limited
LC	Loss Component
LIC	Liability for Incurred Claims
LRC	Liability for Remaining Coverage
MCR	Minimum Capital Requirement
PCR	Prescribed Capital Requirement
RA	Risk Adjustment
RBC	Risk Based Capital
Val Date	The date at which liabilities of an insurer are determined, e.g. final day of a quarter or financial year

1 INTRODUCTION

Purpose

1.1. The IRA issues guidelines to ensure that all insurers and reinsurers operating in Kenya maintain a capital adequacy level commensurate with their risk profile.

1.2. This aims to ensure the financial soundness of insurers and protect policyholders by ensuring that companies can absorb unexpected losses and still meet their obligations.

1.3. This manual explains capital adequacy for insurance companies in Kenya as regulated by the Insurance Regulatory Authority (IRA) for both life and general insurers.

Key Concepts

1.4. **Capital Adequacy:** This refers to the level of capital an insurer must hold to absorb unexpected losses. It is determined by comparing Available Capital against Required Capital.

1.5. **Capital Adequacy Ratio (CAR):** Is the measure of capital available as a percentage (%) of the capital required. An insurer shall maintain a capital adequacy ratio of at least 100% of the MCR.

1.6. **Prescribed Capital Requirement (PCR):** The required level of capital above which the IRA does not take supervisory action. The PCR shall be calculated as 200% of the MCR.

1.7. **Minimum Capital Requirement (MCR):** The minimum statutory level of paid-up capital:

i. Life insurers: KES 400 million

ii. General insurers: KES 600 million

1.8. The IRA's framework is risk-based, meaning companies with riskier profiles need to hold more capital.

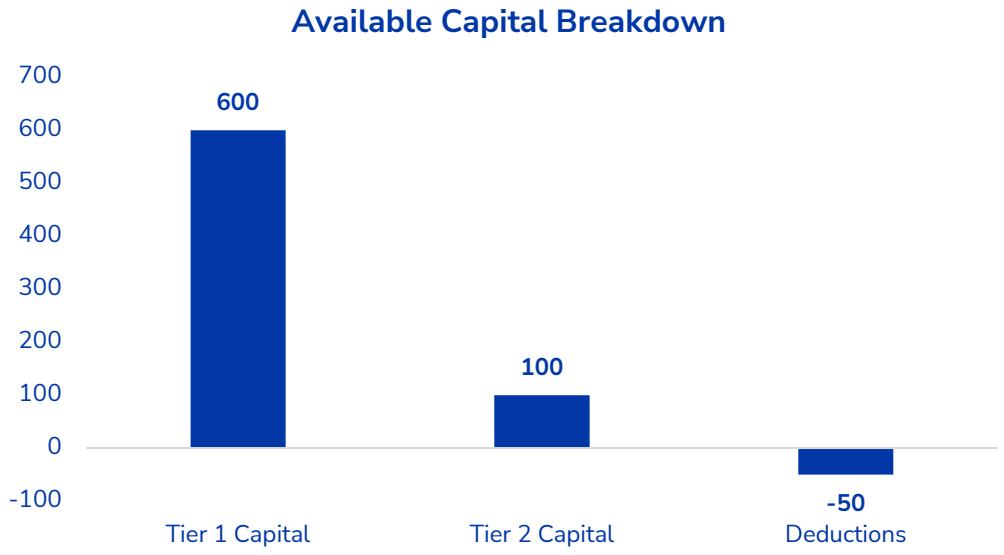
2 CAPITAL ADEQUACY CALCULATION

2.1. Capital Adequacy is computed as below:

$$Capital\ Adequacy = \frac{Total\ Capital\ Available}{Total\ Capital\ Required}$$

Available Capital

2.2. Available Capital is the total amount of capital an insurer has, to cover its risks and is determined as the sum of Tier 1 and Tier 2 capital, less any inadmissible assets.



Tier 1 Capital

2.3. This is the highest quality capital, most loss absorbent and permanent form of capital.

2.4. This shall be an aggregate of the following components:

a) Fully paid-up ordinary shares.

b) Share Premiums.

c) Statutory Reserves.

d) Retained profits/ (accumulated losses).

Tier 2 Capital

2.5. It includes:

a) Irredeemable preference shares.

b) Capital loan stocks or similar instruments.

c) Subordinated loans.

d) Convertible preference shares.

e) Revaluation reserves.

f) Fair value reserves for financial assets.

g) Proposed Dividends.

h) Other reserves.

Deductions (Inadmissible Assets)

2.6. Certain assets are not considered reliable capital resources and must be fully deducted.

2.7. These deductions ensure the reported capital isn't inflated by items that cannot be used to pay claims.

2.8. Deductions include:

a) Goodwill & other intangible assets: Goodwill is an accounting concept representing the value of a company's brand, reputation, etc. It cannot be used to pay claims.

b) The formula used for the deduction is:

$$Max(Sum[intangible\ assets, DAC] - 5\%\ of\ Total\ Assets, 0)$$

Note: DAC (Deferred Acquisition Costs) are costs incurred to acquire new business that are capitalized and expensed over the policy's life.

Goodwill and Intangible Assets	
Intangible Assets	3,900,000
DAC	-
ROU	14,000,000
5% of Total Assets	161,005,000
Goodwill & other intangible assets	-

c) Deferred tax assets.

d) Assets pledged to support credit facilities obtained by an insurer.

e) Other Fixed Assets (including computer equipment, office equipment, motor vehicles and furniture and fittings).

f) Under-reserving.

g) Assets with over concentration limits.

h) Other Deductions (including Prepayments, other receivables, and other assets as determined by the Authority from time to time).

Concentration Limits

2.9. General insurers shall, for the purposes of calculating capital adequacy apply the following concentration limit factors:

i. Deposits in any one financial institution -10% of total assets

ii. Shares of any one institution - 10% of total assets

iii. Property - 30% of total assets

iv. Investment in related parties- 10% of total assets

2.10. Life insurers shall, for the purposes of calculating capital adequacy apply the following concentration limit factors:

i. Deposits in any one financial institution -10% of total assets

ii. Shares of any one institution - 10% of total assets

iii. Property - 50% of total assets

iv. Investment in related parties- 10% of total assets

2.11. The concentration limits for Deposits in financial institutions shall be computed as shown below:

Any one Financial Institution	Fixed Deposit Amount	Limits	Over Conc
Bank 1	150,000,000	322,010,000	-
Bank 2	58,125,000	322,010,000	-
Bank 3	85,000,000	322,010,000	-
Bank 4	20,000,000	322,010,000	-
Bank 5	65,000,000	322,010,000	-
Bank 6	35,000,000	322,010,000	-
Bank 7	25,000,000	322,010,000	-
Bank 8	66,875,000	322,010,000	-
Bank 9	30,000,000	322,010,000	-
	535,000,000		**-**

2.12. The figure below shows a sample computation of over concentration for life insurers:

Total Assets	**3,220,100,000**			
Asset Class	% of Total Assets	Limits	Assets	Over Conc
Property	50%	1,610,050,000	74,000,000	-
Equity	10%	322,010,000	35,000,000	-

Required Capital (Risk Based Capital)

Insurance Risk

2.13. Capital Required for insurance risk aims to cushion the insurer against adverse experience relative to the amount of technical provisions.

General Business

2.14. Insurers writing General Business shall hold capital against fluctuations in the LRC and LIC.

2.15. The capital required by an insurer for insurance risk shall be computed as shown below for each class of business.

$$(Net\ LIC\ \times Risk\ Factor) + (Net\ LRC\ \times Risk\ Factor)$$

2.16. The total LIC reserves used shall be sum of the best estimate liability and the risk adjustment whereas the total LRC shall include the loss component. The LRC and LIC shall be on net basis.

2.17. Risk Factors for each class are provided in the appendix.

Life Business

2.18. The life insurance capital charge is the extra cushion of capital an insurer must hold to cover adverse scenarios.

2.19. It is calculated as the difference between stressed liabilities (using regulator-prescribed stress factors) and the sum of the best estimate liabilities plus the risk adjustment.

2.20. Stress factors are provided in the appendix.

2.21. This shows how much higher liabilities could be if the experience is worse than expected, such as higher deaths, longer lifespans, or higher expenses.

2.22. For Life Insurance business, insurers are required to hold capital to protect against experience resulting from: **mortality, longevity, morbidity/disability, expenses, lapses, and catastrophe risk.**

2.23. The required capital for life insurance risk is calculated using the following formula:

$$\sqrt{Mortality\ risk^2 + Longevity\ risk^2 + Morbidity/Disability^2 + Expenses^2 + Lapses^2 + Catastrophe^2}$$

2.24. For Group Life and Group Credit policies, the risk charges applied to the Liability for Incurred Claims (LIC) and Liability for Remaining Coverage (LRC) are consistent with those prescribed for general insurance business.

2.25. For Deposit Administration and Group Pension Savings plans offered by the insurer, a one percent risk charge shall be applicable to the fund amount.

Catastrophe Risk

2.26. General insurers shall apply a charge of 2% of previous year Net Earned Premium as capital required for catastrophes.

2.27. This shall be added to the insurance risk capital.

Market Risk

2.28. Capital required for market risk aims to cushion against the volatility in the market prices of assets used to back policyholder liabilities.

2.29. Insurers must determine capital for:

i. **Equity Risk:** Exposures to equity instruments including ordinary shares or similar instruments that exhibit market behaviour similar to equities.

ii. **Property Risk:** Exposures to immovable properties both for investment and for self-occupied purposes.

iii. **Interest rate Risk:** Exposures to interest rate related assets and liabilities including debt securities, commercial papers and debentures. The interest rate risk shall be computed for life insurance funds and general insurance funds with discounting factors.

iv. **Currency Risk:** An insurance fund which has exposures in currencies which are different from that of the liabilities will be subjected to a currency risk charge of eight percent on the net open position.

Credit Risk

2.30. This charge cushions against the risk of losses from a counterparty default (e.g., a reinsurer failing to pay a claim)

2.31. An insurer shall, apply risk factors as provided in the appendix.

Operational Risk

2.32. The operational risk capital acts as a cushion against losses that may arise from failed processes, systems and people.

2.33. The Operational Risk Capital shall be computed as the higher of:

i. 30% of the square root of sum of squares of the capital required for insurance risk, market risk and credit risk; and

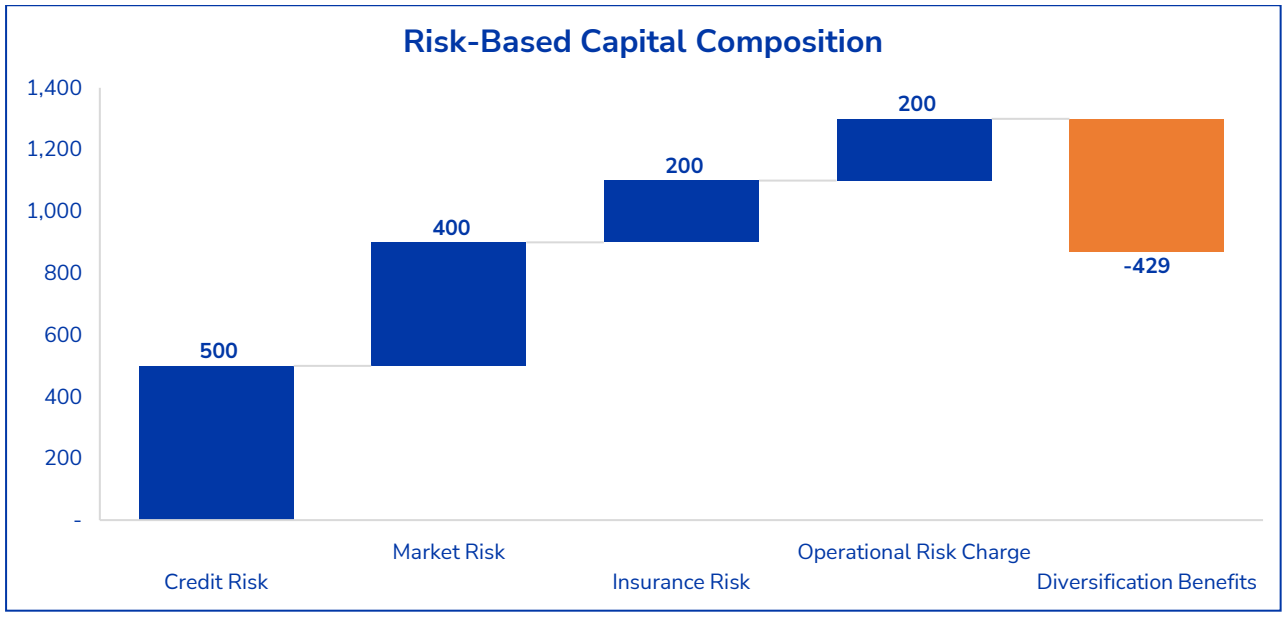
ii. 3% of the gross earned premium over the last 12months.

2.34. The risk-based capital shall be computed as below:

$$\sqrt{Insurance\ Risk\ Capital^2 + Market\ Risk\ Capital^2 + Credit\ Risk\ Capital^2} + Operational\ Risk\ Capital$$

2.35. Diversification benefits represent the reduction in total capital required capital that occurs because risks are not perfectly correlated. This means that when one risk materializes, another may not.

2.36. This reduction allows insurers to hold less capital while remaining solvent under stress.



3 CAR TEMPLATE

3.1. The figure below shows a preview of the template provided for calculation of the CAR for general and life insurers.

Step by step guide to filling the CAR template

a) Set up the Workbook

i. Add a copy of the template for your company and rename it appropriately.

ii. Enter company name and valuation date at the top of the workbook (all tabs should show this correctly).

iii. Add a tab and paste the audited balance sheet. (All tabs should reference this).

b) Total Capital Available Tab (41-1A)

3.2. This tab establishes the capital resources the insurer has available.

i. Enter Tier 1 Capital items.

ii. Enter Tier 2 Capital items.

iii. Enter inadmissible assets to be deducted. (You can use a separate tab to calculate goodwill & other intangible assets).

iv. The template computes Available Capital (Tier 1 plus Tier 2 minus Deductions).

v. Make sure values tie back to the balance sheet.

c) Insurance Risk Tab (41-1B)

3.3. This tab calculates insurance risk charges for the business.

Life business:

i. Enter liabilities (BEL + risk adjustment), net of reinsurance for each product.

ii. Apply the IRA stress factors (mortality, longevity, morbidity, expenses, lapses, catastrophe).

iii. Template derives charges as difference between stressed and best estimate liabilities.

iv. Combine charges using the square-root formula for total life insurance risk.

v. For Deposit Administration and Group Pension Savings, apply a flat 1% charge to the fund amount.

vi. For Group Life and Group Credit, use the same risk factors as general insurance (LIC + LRC).

General business:

vii. Enter LIC and LRC net of reinsurance for each class.

viii. Apply the prescribed LRC and LIC reserve factors to compute charges.

ix. Add catastrophe risk at 2% of prior year Net Earned Premium, then sum to total insurance risk.

d) Market Risk Tab (41-1C)

3.4. Captures risks from assets backing liabilities.

i. Enter balances of asset categories from balance sheet.

ii. Template applies market risk factors to get the risk charge per asset type.

iii. Sum to get total market risk.

e) Credit Risk Tab (41-1D)

3.5. Captures risk of counterparty default.

i. Enter exposures by counterparty (reinsurers, brokers, banks, corporate bonds, loans, receivables).

ii. Apply IRA credit risk factors.

iii. Sum all charges to get total credit risk.

f) Operational Risk Tab (41-1E)

3.6. Captures risks from systems, people, processes.

i. Computes both 30% of the square root of insurance, market and credit risk, and 3% of last 12 months' gross earned premiums.

ii. The template should pick the higher value automatically.

g) RBC Summary Tab (41-1F)

3.7. This tab links totals from insurance risk, Market, credit and operational risks.

i. The risk-based capital is the square root of the sum of the squares of the capital required for insurance, market, and credit risk, plus the capital required for operational risk.

ii. RBC is the required Capital before MCR checks.

h) Capital Adequacy Ratio Tab (41-1)

3.8. This shows the company's capital adequacy ratio.

i. Template computes CAR as Capital Available divided by the Required capital.

ii. Compare to thresholds:

- ≥200%: Comfortable.
- 150–200%: IRA may require reports.
- 110–150%: Submit acceptable remedial plans to IRA.
- Between 110% and MCR: IRA may impose sanctions.
- <100%, breach of MCR: Submit recovery plan/license risk.

3.9. The following tables summarize the components used to calculate capital adequacy for general and life insurers.

General Insurers

RISK BASED CAPITAL - COMPONENTS (in KES millions)		Val Date
A	Credit Risk	X
B	Market Risk	X
C	Insurance Risk	X
D	Operational Risk Charge	X
E	Diversification Benefits	(X)
F = SUM (A .. E)	**Risk Based Capital**	**XX**
REQUIRED CAPITAL (RISK BASED CAPITAL) (in KES millions)		**Val Date**
G	Minimum Capital of KES 600 million	X
H	20% of Preceding year Net Earned Premium	X
I	Calculated Risk Based Capital	X
J= Max (G, H, I)	**Minimum Required Capital**	**XX**
	SOLVENCY RATIO (in KES millions)	**Val Date**
K	Available Capital	X
L	Required Capital	X
M= K/L	**Solvency Ratio / Capital Adequacy Ratio (CAR)**	**XX%**
N = K - L	**At 100% Minimum Capital: Surplus / - Deficit**	XXX
M = K - 2L	**At 200% Minimum Capital: Surplus / - Deficit**	XXX

Life Insurers

RISK BASED CAPITAL - COMPONENTS (in KES millions)		Val Date
A	Credit Risk	X
B	Market Risk	X
C	Insurance Risk	X
D	Operational Risk Charge	X
E	Diversification Benefits	(X)
F = SUM (A .. E)	**Risk Based Capital**	**XX**
REQUIRED CAPITAL (RISK BASED CAPITAL) (in KES millions)		**Val Date**
G	Minimum Capital of KES 400 million	X
H	5% of Insurance Liabilities	X
I	Calculated Risk Based Capital	X
J= Max (G, H, I)	**Minimum Required Capital**	**XX**
	SOLVENCY RATIO (in KES millions)	**Val Date**
K	Available Capital	X
L	Required Capital	X
M= K/L	**Solvency Ratio / Capital Adequacy Ratio (CAR)**	**XX%**
N = K - L	**At 100% Minimum Capital: Surplus / - Deficit**	XXX
M = K - 2L	**At 200% Minimum Capital: Surplus / - Deficit**	XXX

4 SUPERVISORY ACTIONS

4.1. An insurer shall, within thirty days, inform the Authority if the insurer has breached or is likely to breach the prescribed capital requirements.

4.2. A notice by an insurer of the breach or potential breach of these guidelines shall state the remedial action taken or planned to be taken and the period when action shall be taken.

4.3. Where the Authority determines that an insurer has not met the provisions of these guidelines, the Authority may impose any or all of the following administrative sanctions to correct the situation in accordance with the Act, including:

a) require the insurer to invest in a specified manner;

b) restrict or prohibit the insurer from investing in certain asset classes or individual assets to safeguard insurance funds;

c) suspend, dismiss, disqualify or revoke the appointment of an officer of the insurer in a position as a board member, member of the senior management or key person in control function;

d) impose additional reporting requirements on the insurer;

e) restrict declaration and payment of dividends by the insurer;

f) withdraw or impose conditions on the business license of the insurer; and (

g) take any other action as may be deemed necessary.

5 APPENDICES

Appendix 1

a) Insurance Risk Factor- Life Insurance companies

Parameter	Stress Risk Factor
Mortality	6% increase in base mortality rates
Longevity	7% decrease in base mortality rates
Morbidity/Disability - Inception Rates	40% increase in inception rates (Year 1) 15% increase in subsequent morbidity/disability rates 10% decrease in recovery rates
Expenses	5% increase in best estimate expenses
Inflation	1% increase in inflation assumption
Lapses - New Business	15% increase in lapse rates (new business) 5% increase in lapse rates (in-force business)
Interest Rate	10% decrease
Catastrophe	Absolute increase of 1.5 per mille deaths over 1 year

Retirement Savings Business

Parameter	Risk Factor
Pension Plans	1.00%
Deposit Administration	1.00%

b) Insurance Risk Factor- Life Reinsurance companies

Class of Business	Life Reserve Risk Charge
Ordinary life	3.50%
Annuities	1.50%
Investment linked	1.50%

Appendix 2

a) Insurance Risk Factor- General Insurance Companies and Life Insurance Companies transacting Group Life and Group Credit Business

Class of Business	LRC Reserve Risk Charge	LIC Reserve Risk Charge
Aviation	39%	29%
Engineering	8%	4%
Fire domestic	3%	2%
Liability	9%	9%
Marine	7%	8%
Motor vehicle private—Property damage	5%	5%
Motor vehicle private—Liability	12%	12%
Motor vehicle commercial—Property damage	3%	3%
Motor vehicle commercial—Liability	13%	13%
Motor vehicle commercial PSV—Property damage	3%	3%
Motor vehicle commercial PSV—Liability	14%	14%
Personal accident	6%	9%
Theft	4%	4%
Workmen's compensation	18%	19%
Miscellaneous	8%	6%
Medical	15%	13%
Group life	10%	8%
Group credit	12%	10%

b) Insurance Risk Factor- Reinsurance Companies

Class of Business	LRC Reserve Risk Charge	LIC Reserve Risk Charge
Aviation	42%	26%
Engineering	9%	6%
Fire domestic	5%	3%
Liability	13%	12%
Marine	12%	10%
Motor vehicle private—Property damage	8%	7%
Motor vehicle private—Liability	16%	15%
Motor vehicle commercial—Property damage	6%	5%
Motor vehicle commercial—Liability	17%	16%
Motor vehicle commercial PSV—Property damage	5%	5%
Motor vehicle commercial PSV—Liability	17%	16%
Personal accident	10%	12%
Theft	7%	6%
Workmen's compensation	20%	22%
Miscellaneous	8%	8%
Medical	15%	13%
Group life	10%	8%
Group credit	12%	10%

Appendix 3

a) Credit risk – Insurance and Reinsurance Companies Capital Charges

Asset	Risk Charge
Government securities—National and County	0%
Foreign government bonds	5%
Corporations' and other organizations' bonds	12%
Term deposits	0%
Cash and cash balances	0%
Policy loans	0%
Secured loans—corporations and other organizations	10%
Secured loans—staff and individuals	30%
Investment in subsidiaries, associates and joint ventures	40%
Mortgages	30%
Secured loans to related parties	100%
Category 1—Reinsurers rated above A-	1.50%
Category 2— Reinsurers rated above BBB	10%
Category 3— Reinsurers rated below BBB	35%
Category 4— Reinsurers unrated	100%
Category 5 – Reinsurers licensed under the Insurance Act	2.50%
Insurance receivables—amount outstanding	100%
Reinsurance receivables—amount outstanding	100%
Assets under Deposit Administration and Pension Plans	1.50%
Assets under Unit Linked	2.00%

Appendix 4

Market risk – Insurance and Reinsurance Companies Capital Charges

Equity Type	Risk Charge
Listed ordinary shares on NSE	30%
Listed ordinary shares on regional exchanges	30%
Listed preference shares on NSE	35%
Listed preference shares on regional exchanges	40%
Unlisted shares or private equity	40%

Interest rate risk

Scenario	Asset value	Liability value	Surplus
Base interest rate			
Increasing interest rate			
Decreasing interest rate			
Interest rate Risk Charge			

Currency risk

Type of currency	Net on balance sheet position	Net forward position	Net long position	Net short position
Total position				
Exposure				
Currency risk charge				8%
Currency risk capital charge				

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About Us

Kenbright Actuarial & Financial Services Limited (KAFS)
Prism Towers, 26th Floor
3rd Ngong Avenue, Nairobi
Kenya
www.kenbright.co.ke

Customer Service Lines: +254 (020) 22 111 52

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Key Contacts

E Macharia, FIA C.Act, FeASK
Chief Actuary & Managing Director
E-mail: ezekiel@kenbright.africa
Phone: +254 722 540 045